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Digital Twin
CHALLENGE
Build. Predict. Reimagine Healthcare.

REIMAGINING & REFORMING
HEALTHCARE IN INDIA

Digital Twin Challenge 2026

- **Problem Statement Title -** Type 2 Diabetes Predicting post-meal glucose spikes up to 2 hours ahead

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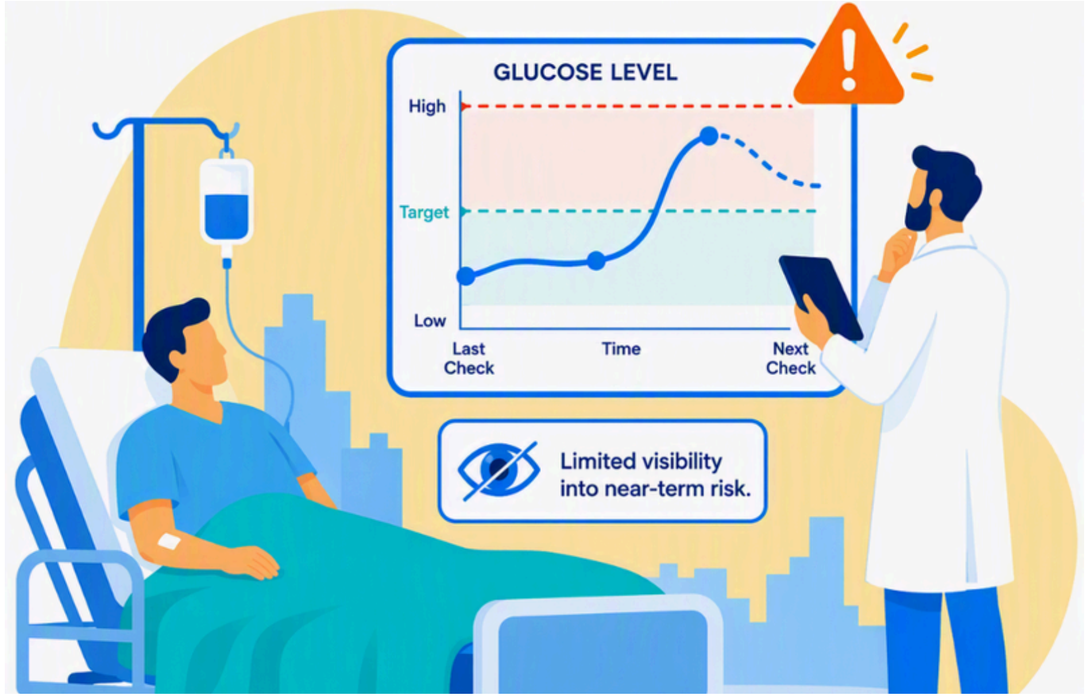
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Detailed problem statement

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The problem (Type 2 Glucose): glucose spikes are easy to miss

A patient's glucose can rise between routine checks, leaving clinicians with limited visibility into near-term risk.



Hidden in time

A post-meal rise may happen well before the next appointment or manual check.

Multiple signals

Glucose, recent meals, activity and heart rate can all provide useful context.

Hard to interpret

A raw timeline shows what happened; clinicians also need an understandable warning about what may happen next.

PROJECT GOAL

Estimate whether glucose will cross 180 mg/dL within the next 2 hours, then present the risk and its likely drivers in a clinician-oriented dashboard.

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Surveys

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Why India Needs Smarter Diabetes Monitoring

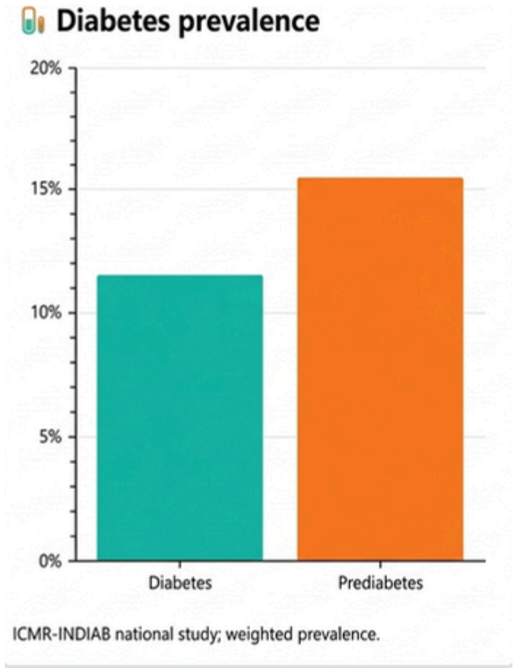
Evidence from nationally representative Indian studies

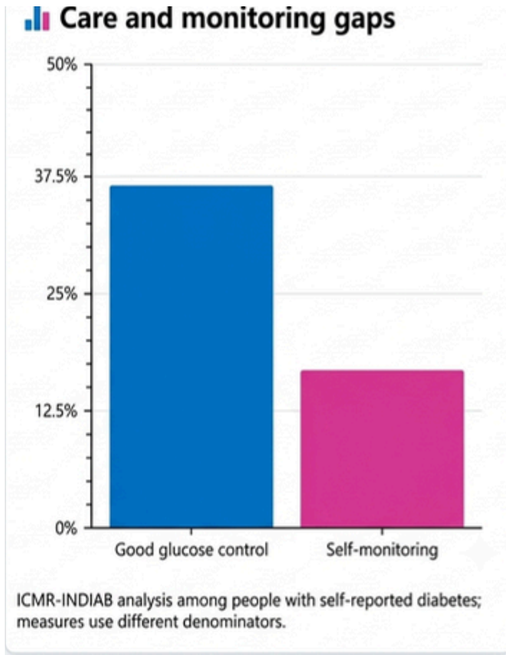
11.4%
Diabetes prevalence
Weighted prevalence among Indian adults in the ICMR-INDIAB study.

15.3%
Prediabetes prevalence
Shows the wider population at risk of metabolic disease.

36.3%
Good glucose control
Among participants with self-reported diabetes in a separate ICMR-INDIAB analysis.

16.7%
Reported glucose self-monitoring
Among the study population with self-reported diabetes.





These findings come from two analyses of the ICMR-INDIAB study. The first assessed diabetes prevalence in 113,043 participants; the second examined diabetes treatment targets and monitoring practices.

- **ICMR-INDIAB national survey (2023)**
- **National NCD Monitoring Survey analysis**
- **ICMR-INDIAB diabetes treatment study (2022)**

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Proposed technical solution

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Frontend

Streamlit

matplotlib

Backend

NumPy

pandas

Shap

XGBoost

Joblib

Data and Authentication

SQLite

ReportLab

Firebase

A patient-specific digital representation combines a clinical profile with recent sensor data.

PATIENT DATA

Clinical profile & wearable timeline

PREDICTION

2-hour glucose-spike risk estimate

CLINICIAN ACTION

Risk level, reasons & meal what-if

Spike warning

Low / Watch / High with plain-language reasons.

Indian meal what-if

Compare estimated peak glucose for selected dishes and portions.

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Conceptual architecture

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Streamlit

matplotlib

User Dashboard

User Input & Interaction

Patient Data View

Digital Twin Insights

Data Cleaning

Feature Engineering

XGBoost

Model Training

Shap

CGM Data

Patient Profile

.CSV

PKL

JSON

Indian Meals Dataset

Research Dataset

Trained Models

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Working principle

1. Data Collection

Patient profile, historical glucose readings, CGM data and lifestyle information

2. Data Processing

Clean data, extract useful features and prepare model inputs

3. AI Prediction

Machine-learning models estimate the risk of glucose exceeding 180 mg/dL within the next 2 hours

4. Digital Twin Simulation

Simulate virtual patient scenarios and compare possible meal-related glucose responses

5. Dashboard & Insights

Display predicted risk, glucose trends and model explanations through a Streamlit dashboard

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Estimated cost

COMPONENT	ESTIMATED COST
Python, Streamlit, Scikit-learn, XGBoost	Rupees 0 for initial prototype
Public / simulated datasets	Rupees 0 (OpenSource)
Model training and visualization	Rupees 0 for initial prototype and small dataset
Database(SQL)	Rupees 0 initially
Local development and testing	Rupees 0-2000
Hosting / deployment	Rupees 0-1000

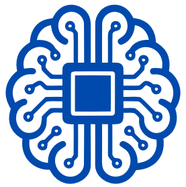
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Team composition and roles



Member 1
Data & Research
Dataset collection, data cleaning, feature selection and research on diabetes.



Member 2
AI/ML Development
Train prediction models, evaluate performance and implement explainability.



Member 3
Frontend & Digital Twin
Build the Streamlit dashboard, patient simulation and meal what-if interface.

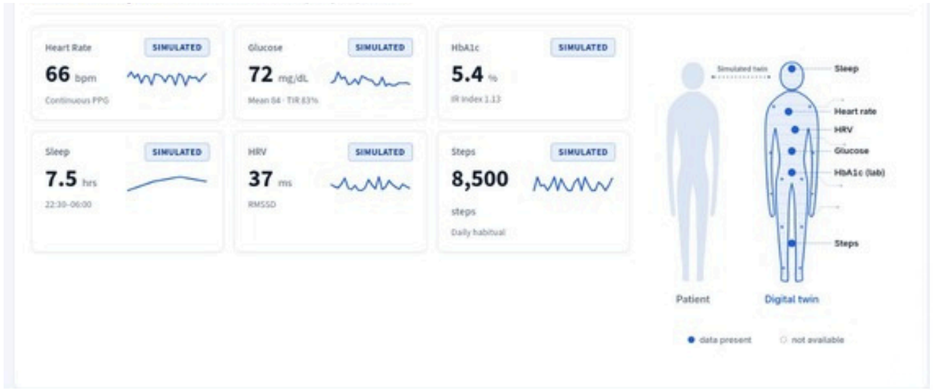


Member 4
Integration & Testing
Connect modules, test the complete system, manage documentation and presentation.

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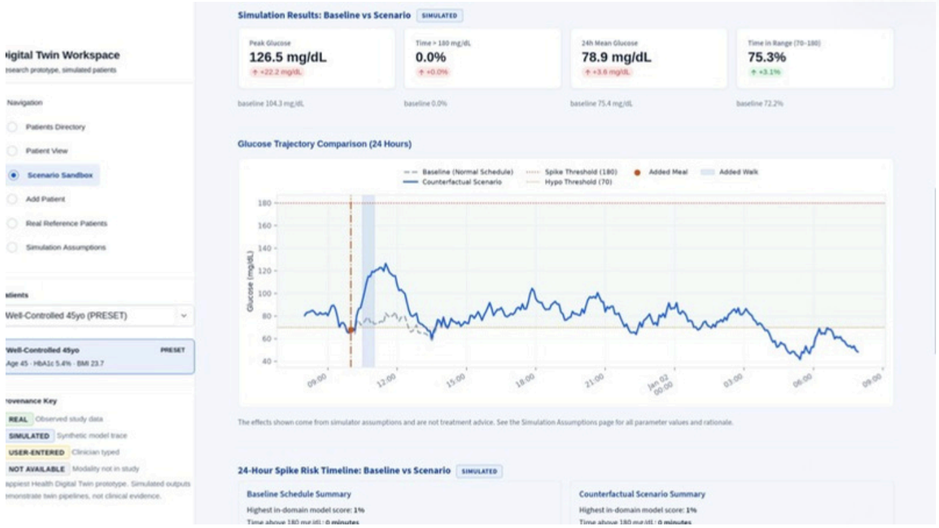
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Prototype Design











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